

MES Maya

Project Overview

MES Maya is an AI-powered voice assistant designed for hospitals and clinics. It acts as a virtual receptionist, handling patient calls by providing doctor availability, consultation schedules, fees, and intelligent call transfers. The system integrates FreeSwitch PBX, Google Vertex AI Gemini Live, and Firebase Firestore to deliver real-time, low-latency voice interactions.

Key Features

- AI-based virtual receptionist
- Real-time voice conversations
- Doctor availability and consultation details
- Intelligent call routing
- Multi-tenant hospital support
- Dynamic AI prompts
- Voice Activity Detection (VAD)
- Audio processing and noise reduction
- FastAPI dashboard APIs
- Centralized logging and monitoring

Technologies Used

- Python 3.9+
- FastAPI
- WebSockets
- Google Vertex AI (Gemini Live)
- Firebase Firestore
- FreeSwitch PBX
- NumPy & SciPy

Conclusion

MES Maya combines AI, telephony, and cloud technologies to automate hospital reception services. It provides accurate, real-time patient assistance while reducing manual workload, making it a scalable and efficient solution for modern healthcare communication.